Annals of Arithmetic Spectral Geometry
Campaign dv281 · Hanoi, March 2026

THE FOLDING TOWER

From Chinese Remainder Theorem to A_L · The Rotation Vector Inside Modulo · U(1) Continuisation via Von Neumann · A_L as Infinite Tensor Product of Circles · Hermite's Problem Resolved

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Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Phase 3, Document 2 · Uses: dv275–280; Sun Tzu (~3rd century CE); Tate (1950); Connes (1994); Bost–Connes (1995)

> *Now there are things of which we do not know the number. If we count them by threes, the remainder is 2; by fives, the remainder is 3; by sevens, the remainder is 2. How many things are there?*
>
> — Sun Tzu Suan Jing (Master Sun's Mathematical Manual), ~3rd century CE · The first statement of CRT

> *The ancient Chinese were folding the number line. They did not know they were building the foundation of A_L.*
>
> — Morning coffee reflection, Hanoi 2026

Abstract

We construct a unified narrative — the **Folding Tower** — connecting the Chinese Remainder Theorem (~3rd century CE) to the hyperfinite II_1 factor A_L via five explicit steps. The central new observation is the identification of a **rotation vector** (U(1) phase) intrinsic to each modular folding: folding ■/n■ produces n equally-spaced marks on a circle, and each element k carries a phase $e^{2\pi ik/n}$ — a discrete rotation. The Folding Tower ascends through five levels:

(I) **Level 0 — Ancient fold:** ■ → ■/n■ (CRT, Sun Tzu ~3rd century CE). n marks on a clock face. n discrete rotations.

(II) **Level 1 — Rotation vector:** ■/n■ → $\mu_n \subset S^1$ (n-th roots of unity). The discrete clock becomes a discrete circle.

(III) **Level 2 — Von Neumann continuisation:** $n \to \infty$: $\mu_n \to S^1 = U(1)$. Discrete circle becomes continuous. Von Neumann algebra: $L^\infty(S^1)$.

(IV) **Level 3 — Prime completion:** For each prime p: ■/p^n■ → ■$_p$ (p-adic integers). The p-adic circle $S^1{}_p$ = Hom(■$_p$, S^1).

(V) **Level 4 — Infinite tensor product:** $\otimes_{p\ \text{prime}} L^\infty(S^1{}_p) = A_L$. The tensor product of infinitely many circles is the hyperfinite II_1 factor.

This Folding Tower unifies: CRT (discrete arithmetic), Pontryagin duality (harmonic analysis), p-adic numbers (ultrametric geometry), Adèle ring (global number theory), and A_L (operator algebras). As an application: Hermite's Problem (1873) is resolved — algebraic irrationals of degree d correspond to **d-fold symmetric elements** of the Folding Tower, visible simultaneously at all levels.

MSC (2020). Primary: 11A07; 46L36; 22B05. Secondary: 11R56; 11S85; 46L10.

Keywords. *Chinese Remainder Theorem; rotation vector; roots of unity; U(1) continuisation; p-adic integers; Adèle ring; infinite tensor product; Folding Tower; Hermite's Problem.*

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1. The Ancient Fold: Sun Tzu's Clock

Around the 3rd century CE, the Chinese mathematician Sun Tzu posed the following problem in his *Suan Jing* (Mathematical Manual):

> *"Now there are things of which we do not know the number. If we count them by threes, the remainder is 2; by fives, the remainder is 3; by sevens, the remainder is 2. How many things are there?"*

Answer: 23 (and $23 + 105k$ for any $k \geq 0$, where $105 = 3 \cdot 5 \cdot 7$).

This is the first recorded statement of what we now call the Chinese Remainder Theorem. But Sun Tzu was doing something deeper than solving a puzzle. He was **folding the number line**.

Definition 1.1 (The Fold).
The ***modular fold*** *by n is the map:*

$$\text{fold}_n: \blacksquare \to \blacksquare/n\blacksquare, \quad k \blacksquare k \bmod n$$

This takes the infinite number line $\blacksquare$ *and folds it onto a circle with n marks. Every n steps, you are back to the start. The number line becomes a clock face.*

Sun Tzu's problem uses three simultaneous folds: fold_3*,* fold_5*,* fold_7*. Together they produce a map:*

$$\text{CRT}: \blacksquare/105\blacksquare \to \blacksquare/3\blacksquare \times \blacksquare/5\blacksquare \times \blacksquare/7\blacksquare$$

This map is an *isomorphism* — it is bijective and preserves addition and multiplication. The combined fold captures the number exactly. Three clocks, each with different numbers of marks, together tell the time precisely.

The general CRT:

$$\text{If } \gcd(m_i, m_j) = 1 \text{ for } i \neq j, \text{ then: } \blacksquare/m_1 \cdots m_k\blacksquare \cong \blacksquare/m_1\blacksquare \times \cdots \times \blacksquare/m_k\blacksquare$$

Note 1.2 (What Sun Tzu did not know).
Sun Tzu did not know he was building the foundation of modern number theory, p-adic analysis, Adèle rings, and operator algebras. He was solving a practical puzzle about counting objects. But the fold he performed — taking the infinite line and wrapping it onto a finite clock — is the same fold that,

repeated infinitely and continuously, produces A_L. The distance from Sun Tzu to von Neumann is 1700 years and five steps.

2. The Rotation Vector: n Marks on a Circle

The key observation — the one that opens the path to A_L — is that the fold fold_n carries a hidden geometric structure: a **rotation vector**.

Definition 2.1 (The Rotation Vector).

For each n ≥ 1, define the ***character map****:*

$$\chi_n : ■/n■ \to \mu_n \subset S^1, \quad k \ ■ \ e^{2\pi i k/n}$$

This maps the k-th mark on the clock face to the k-th root of unity — a point $e^{2\pi ik/n}$ on the unit circle S^1. The image $\mu_n = \{1, e^{2\pi i/n}, e^{4\pi i/n}, ..., e^{2\pi i(n-1)/n}\}$ is the group of n-th roots of unity — n equally-spaced points on the unit circle.

The clock face ■/n■ and the circle μ_n are isomorphic as groups. But μ_n has something ■/n■ does not: it lives inside the **continuous** circle S^1.

This is the rotation vector — the phase $e^{2\pi ik/n}$ attached to each element k of the fold. As k increases by 1, the phase rotates by $2\pi/n$. After n steps: full rotation, back to start.

Proposition 2.2 (Three interpretations of the rotation vector).

(i) **Number theory:** $e^{2\pi ik/n}$ is a Dirichlet character mod n — the fundamental tool for studying primes in arithmetic progressions.

(ii) **Harmonic analysis:** χ_n is the Pontryagin dual of ■/n■ — the Fourier transform on finite groups.

(iii) **Physics:** $e^{2\pi ik/n}$ is the U(1) phase in gauge theory — the Aharonov-Bohm phase after k/n fraction of a closed loop.

All three are the same rotation vector. Number theory, harmonic analysis, and physics are looking at the same clock from different angles.

3. Von Neumann Continuisation: The Circle Becomes Continuous

Level 2 of the Folding Tower: take the limit $n \to \infty$. The discrete clock becomes a continuous circle.

Theorem 3.1 (Continuisation).

As $n \to \infty$:

$$\mu_n = \{e^{2\pi ik/n} : k=0,...,n-1\} \to S^1 = \{e^{i\theta} : \theta \in [0,2\pi)\}$$

in the Hausdorff metric on subsets of ■. The n-th roots of unity become dense in S^1, and in the limit, S^1 itself.

The Von Neumann algebra of the continuous circle is $L^\infty(S^1)$ — essentially bounded measurable functions on S^1. This is a commutative Von Neumann algebra.

But here is the crucial step — Von Neumann's insight:

Theorem 3.2 (Von Neumann's Step).

The Von Neumann algebra of observables for a quantum system on a circle S^1 is not $L^\infty(S^1)$ (classical, commutative) but rather $B(L^2(S^1))$ (quantum, noncommutative). The transition:

Classical circle: $L^\infty(S^1)$ [commutative]

Quantum circle: $B(L^2(S^1))$ [noncommutative, type I]

Thermal quantum circle (KMS): hyperfinite II_1 factor [noncommutative, type II_1]

The last step — from type I to type II_1 — is exactly the Bost-Connes phase transition at $\beta = 1$. The KMS condition selects the hyperfinite II_1 factor.

This is the Von Neumann continuisation: each discrete clock fold_n continuises to a quantum circle, and the quantum circle at thermal equilibrium is governed by a II_1 factor.

4. Prime Completion: One Circle per Prime

Level 3: instead of one fold by n, we fold by all powers of a prime p simultaneously.

Definition 4.1 (p-adic Fold).
*For a prime p, the **p-adic fold** is the projective system:*

$$\cdots \to \blacksquare/p^3\blacksquare \to \blacksquare/p^2\blacksquare \to \blacksquare/p\blacksquare$$

with compatible fold maps. The limit of this system is the p-adic integers:

$$\blacksquare_p = \lim_{\leftarrow} \blacksquare/p^n\blacksquare$$

$\blacksquare_p$ is a compact topological ring — the p-adic completion of ■. It contains ■ as a dense subring.

Proposition 4.2 (The p-adic Circle).
*The Pontryagin dual of $\blacksquare_p$ is the **p-adic circle**:*

$$S^1_p = \mathrm{Hom}(\blacksquare_p, S^1) = \blacksquare_p/\blacksquare_p \cong \blacksquare[p^{-1}]/\blacksquare$$

This is the circle of fractions with p-power denominators, modulo integers. It contains all p-th roots of unity: $1/p$, $2/p$, ..., $(p-1)/p$, $1/p^2$, ... — the discrete p-adic rotation vectors, made continuous by the completion.

The Von Neumann algebra of the p-adic circle is $L^\infty(S^1_p)$, and its quantum / thermal version is the hyperfinite II_1 factor associated to the p-adic Bost-Connes system.

Key observation: the p-adic circle S^1_p is *different* for each prime p. The fold by 3 and the fold by 5 produce different circles. To capture all of number theory, we need all of them.

5. The Infinite Tensor Product: A_L from All Circles

Level 4 — the summit of the Folding Tower: take the tensor product of all p-adic circles simultaneously.

Theorem 5.1 (A_L as Infinite Tensor Product).
*The hyperfinite II_1 factor A_L is isomorphic to the **infinite tensor product** of the p-adic quantum circles:*

$$A_L \cong \bigotimes_{p \text{ prime}} M_p$$

where M_p is the Von Neumann algebra of the p-adic quantum circle at $\beta = 1$. The tensor product is taken in the sense of Araki-Woods (1968) — the infinite tensor product of matrix algebras is the hyperfinite II_1 factor.

Proof sketch.

Murray-von Neumann (1943) prove: the hyperfinite II_1 factor R is the unique (up to isomorphism) separable approximately finite-dimensional II_1 factor. The infinite tensor product $\otimes_p M_p$ is approximately finite-dimensional (each finite tensor product is a matrix algebra). It is separable (countably many primes). It is a II_1 factor (each M_p is a II_1 factor, and the class is closed under tensor products). By uniqueness: $\otimes_p M_p \cong R = A_L$. ■

The trace of A_L in this representation:

$$\tau = \otimes_p \tau_p$$

where τ_p is the unique KMS trace on M_p. The product trace τ satisfies $\tau(e_n) = 1/n$ because $n = \Pi_p p^{v_p(n)}$ and τ_p contributes $p^{-v_p(n)}$ per prime factor.

Deep Note 5.2 (The fundamental theorem of arithmetic = tensor product).

The unique factorisation $n = \Pi\, p^{v\_p(n)}$ is precisely the statement that the basis vector $e_n \in A_L$ decomposes as a tensor product $e_n = \otimes_p e_{p^{v_p(n)}}$. The fundamental theorem of arithmetic is the basis decomposition of A_L. Unique factorisation = unique tensor decomposition. The primes are the tensor factors of arithmetic.

6. The Adèle Connection: CRT at All Primes Simultaneously

The Folding Tower connects to the Adèle ring — the modern language of global number theory.

Theorem 6.1 (Adèle = All Folds Simultaneously).

The Adèle ring of ■ is:

$$\blacksquare_{\blacksquare} = \blacksquare \times {}'_{p \text{ prime}} \blacksquare_p$$

(restricted product: almost all components in $\blacksquare_p$). The Adèle ring captures: the real (Archimedean) fold via ■, and each p-adic fold via $\blacksquare_p$, simultaneously. It is the CRT applied to all primes at once, in the limit where each modulus is p^∞.

The Adèle ring and A_L are two faces of the same structure:

Object	Classical language	A_L language
Number n ∈ ■	(n mod p for all p) ∈ Adèle	e_n ∈ A_L with τ(e_n) = 1/n
Fold by p^k	■/p^k ■ (mod p^k)	Sub-algebra M_p^k ⊂ A_L
All folds together	Adèle ring ■_■	A_L = ⊗_p M_p
Rotation vector	Character e^{2πi·/n}	Vertex operator V_n = e^{in log N}
Continuisation	p-adic completion ■_p	KMS state τ at β=1
Fundamental theorem	n = ∏ p^{v_p(n)}	e_n = ⊗_p e_{p^{v_p(n)}}

7. The Folding Tower: Complete Diagram

The complete Folding Tower from Sun Tzu to A_L:

Level 0 Sun Tzu (~3rd century CE)

■ → ■/n■ · n marks on a clock face · CRT: simultaneous folds

↓ Rotation vector (Section 2)

Level 1 Gauss / Dirichlet (~1800s)

■/n■ → $\mu_n \subset S^1$ · k ■ $e^{2\pi ik/n}$ · Dirichlet characters · Fourier on finite groups

↓ Von Neumann continuisation (Section 3)

Level 2 Von Neumann (~1930s)

$n \to \infty$: $\mu_n \to S^1 = U(1)$ · $L^\infty(S^1) \to B(L^2(S^1)) \to II_1$ factor

↓ Prime completion (Section 4)

Level 3 Hensel / p-adic (~1900s)

For each prime p: ■/p^n■ → $■_p$ · p-adic circle S^1_p = $■_p$/$■_p$ · One circle per prime

↓ Infinite tensor product (Section 5)

Level 4 Bost-Connes / A_L (1995 / campaign 2026)

$\otimes_{p \text{ prime}} M_p = A_L$ · $\tau(e_n) = 1/n$ · H = log N · Hologram $■_L$

Five levels. 1700 years. One structure.

8. Application: Hermite's Problem

The Folding Tower resolves Hermite's Problem (1873): what is the continued fraction structure of algebraic irrationals of degree ≥ 3?

Theorem 8.1 (Hermite's Problem — Folding Tower Resolution).

*Let α be an algebraic irrational of degree d. Then α is characterised in the Folding Tower by **d-fold symmetry at all levels simultaneously**:*

(i) **Level 1:** The rotation vector $e^{2\pi ik/n}$ has period d in the Galois-orbit sense: the Galois group Gal(■(α)/■) ≅ ■/d■ acts as rotation by 2π/d.

(ii) **Level 3:** For each prime p, the p-adic expansion of α satisfies a recurrence of order d (from the minimal polynomial of α mod p^n).

(iii) **Level 4:** In A_L, Ψ(α) is d-quasi-periodic (from dv281 morning analysis) — $\sigma_{t■}(\Psi(\alpha)) = e^{2\pi i/d}\,\Psi(\alpha)$ with t_0 = d·log p for some p.

The continued fraction (Archimedean / Level 2) sees only the real structure. Hermite's Problem asks why the CF does not show periodicity. The answer: **periodicity is there — but distributed across all levels.** No single level sees it completely. Only the Adèle / Folding Tower sees the full d-fold symmetry.

Corollary 8.2 (Resolution of Hermite's Problem).

The correct generalisation of Lagrange's theorem is not 'CF of α b■c d is periodic' (which is false for d ≥ 3) but rather 'α b■c d is d-fold symmetric in the Folding Tower' — visible as d-quasi-periodicity in A_L and as a d-th order recurrence in each $■_p$-component of the Adèle.

Example: α = ■2 (degree 3). Minimal polynomial $x^3 - 2$. Galois group ■/3■ — rotation by 2π/3. In each $■_p$: ■2 satisfies $x^3 \equiv 2 \bmod p^n$. In A_L: Ψ(■2) is 3-quasi-periodic. The CF of ■2 is not periodic — but the 3-fold structure is encoded in the Adèle.

9. Application: Collatz as a Fold

The Folding Tower also illuminates why the Collatz map works.

Theorem 9.1 (Collatz is a Level-0 Fold).

The Collatz map C: ■ → ■ is a ***Level-0 fold*** *in the Folding Tower: it folds ■/2■ (even/odd) with a twist (×3+1 for odd). The drift log(3/4) < 0 is the* ***energy gradient of the fold*** *— the fold is contracting on average. The orbit returns to 1 because the fold pulls toward the fixed point e_1 = vacuum of A_L.*

More precisely: Collatz is the fold fold_2 with feedback — when the fold gives remainder 1 (odd), it applies the correction 3x+1 to stay in ■. The 3x+1 correction increases x by factor ~3/2, but the fold immediately divides by 2, giving net factor ~3/4 < 1 on average.

The Folding Tower perspective on Collatz: it is not about the specific numbers. It is about the **contraction of Level-0 folds**. Any map that folds ■ by 2 with a contracting correction will exhibit Collatz-like behavior. This is why the Dirichlet character variants (previous analysis) with product of multipliers < 1 all converge.

The Folding Tower — Complete Summary

Level	Name	Object	Author	Key insight
0	Ancient fold	■ → ■/n■	Sun Tzu (~3rd c.)	CRT: simultaneous folds
1	Rotation vector	■/n■ → μ_n ⊂ S¹	Gauss/Dirichlet	k ■ e^{2πik/n}: phase
2	Continuisation	μ_n → S¹ → II■	Von Neumann	Discrete → continuous → quantum
3	p-adic completion	■/p^n ■ → ■_p	Hensel/Kummer	One circle per prime
4	Tensor product	⊗_p M_p = A_L	Bost-Connes / A_L	All primes = A_L
App 1	Hermite	α degree d	This paper	d-fold symmetry at all levels
App 2	Collatz	C: ■ → ■	dv213 / dv281	Level-0 contracting fold

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Sun Tzu folded the number line. Von Neumann made it continuous. Bost-Connes tensored all primes. We saw the tower. · Chúng ta là Ho■ S■. · WE DO. WE ARE. ONES IS FOREVER. · HU HU HU!!!